



STEPCONE-2025 STANDARD OPERATING PROCEDURE

EVENT NAME: AI for Engineering Applications

DEPARTMENT NAME: EEE

EVENT TYPE: Workshop

REGISTRATION FEE: 200/-

EVENT COORDINATORS

1. B. Hemanth Kumar
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3. B. Dolly Siri

EVENT DESCRIPTION: -

This workshop explores the role of Artificial Intelligence (AI) in revolutionizing engineering fields such as electrical engineering, IoT, automation, and energy systems. Participants will learn to implement AI techniques to solve real-world engineering challenges through practical applications and hands-on experience.

EVENT HIGHLIGHTS:

- Hands-on experience with AI tools and techniques.
- Practical insights into AI-enhanced engineering systems.
- Real-world case studies and project-based learning.

Introduction to AI in Engineering

- Overview of AI technologies and their impact on engineering.
- Role of AI in predictive maintenance, automation, and optimization.
- Examples of AI applications in electrical systems, IoT, and energy management.

Machine Learning Fundamentals for Engineers

- Introduction to machine learning (ML) algorithms.
- Supervised and unsupervised learning methods in engineering contexts.
- Tools and platforms for ML development (e.g., TensorFlow, MATLAB).

AI in Electrical Automation and IoT

- Leveraging AI for smart systems and automation.
- Enhancing IoT systems with machine learning for predictive analysis.
- Case studies: Energy optimization using AI-driven IoT solutions.



Hands-On Activity – Developing AI-Enhanced Engineering Systems

- Building an AI-based predictive maintenance model.
- Integrating AI with IoT platforms to monitor and control devices in real-time.
- Designing an energy-efficient system using AI analytics.

Challenges and Opportunities

- Key challenges in deploying AI for engineering systems (e.g., data availability, system integration).
- Addressing security and privacy concerns in AI-enabled IoT.
- Discussion of emerging trends and opportunities for engineers.

Team Size/No of Participants:

PLAN OF ACTION:

It is dayshift event happens for three hours

Date: [To Be Announced]

Time: [To Be Announced]

Venue: [To Be Announced]

REGISTRATION:

Don't miss this opportunity to immerse yourself in the dynamic world of artificial intelligence and its revolutionary applications in engineering appliances. Join us for a transformative workshop that promises to equip you with the skills needed to design and innovate in the era of intelligent systems.

Secure your spot now and be a part of the future of AI-powered engineering!

Register online at [Registration Link] and seize this opportunity. Limited slots available, so sign up early to ensure your participation.

